

CareerOps: A Multi-Agent Orchestration Framework for Automated Resume Optimization and Application Tracking

Mr. Prashanth Vadityavath
Ms in Data Analytics and
Visualization
Yeshiva University, NY
pvaditya@mail.yu.edu

Mr. Karthik Reddy Gaddam
Seshadri
Ms in Data Analytics and
Visualization
Yeshiva University, NY
kgaddams@mail.yu.edu

Mr. Ganesh kumar Soma
Ms in Data Analytics and
Visualization
Yeshiva University, NY
gsoma@mail.yu.edu

Mr. Brandon Chiazza
Ms in Data Analytics and Visualization
Yeshiva University, NY
brandon.chiazza@yu.edu

Abstract

The modern job search is characterized by the need for highly tailored application materials, a process that is traditionally manual, repetitive, and time-consuming. While Large Language Models (LLMs) offer generative capabilities, isolated interactions lack the persistence and context required for a continuous job-search workflow. This paper introduces CareerOps, an integrated, AI-powered career operating system. By combining a robust PostgreSQL data foundation with a LangChain-driven multi-agent architecture, CareerOps automates resume analysis, continuous tailoring, and application tracking. We detail the system's modular architecture, which supports both proprietary cloud APIs and local open-weight models, and present preliminary statistical evaluations comparing model latency, schema adherence, and optimization efficacy.

Keywords: Multi-Agent Systems, Large Language Models, Generative AI, Applicant Tracking Systems, Workflow Automation, LangChain.

1. Introduction

The transition from broad job applications to highly targeted, keyword-optimized submissions has created a significant bottleneck for job seekers. Candidates are frequently required to parse complex job descriptions and manually map their experiences to specific requirements to bypass automated Applicant Tracking Systems (ATS) (Woods et al., 2020).

While candidates increasingly utilize generative AI to assist in this process, standard zero-shot prompting in isolated web interfaces leads to inconsistent outputs and a loss of contextual memory (Bubeck et al., 2023). Users are forced into a repetitive loop of copying and pasting base resumes and job descriptions, with no centralized system to track which generated version corresponds to which application.

CareerOps addresses this gap by proposing a unified platform that bridges data persistence, AI generation, and workflow analytics. By orchestrating a specialized team of AI agents each constrained to a specific analytical or generative task CareerOps acts as an autonomous career command center, reducing manual friction

and improving the systematic tracking of application success.

frontend interface, an intelligence layer, and a persistent data foundation.

2. Methodology and System Architecture

CareerOps is designed as a modular, three-tier application encompassing a

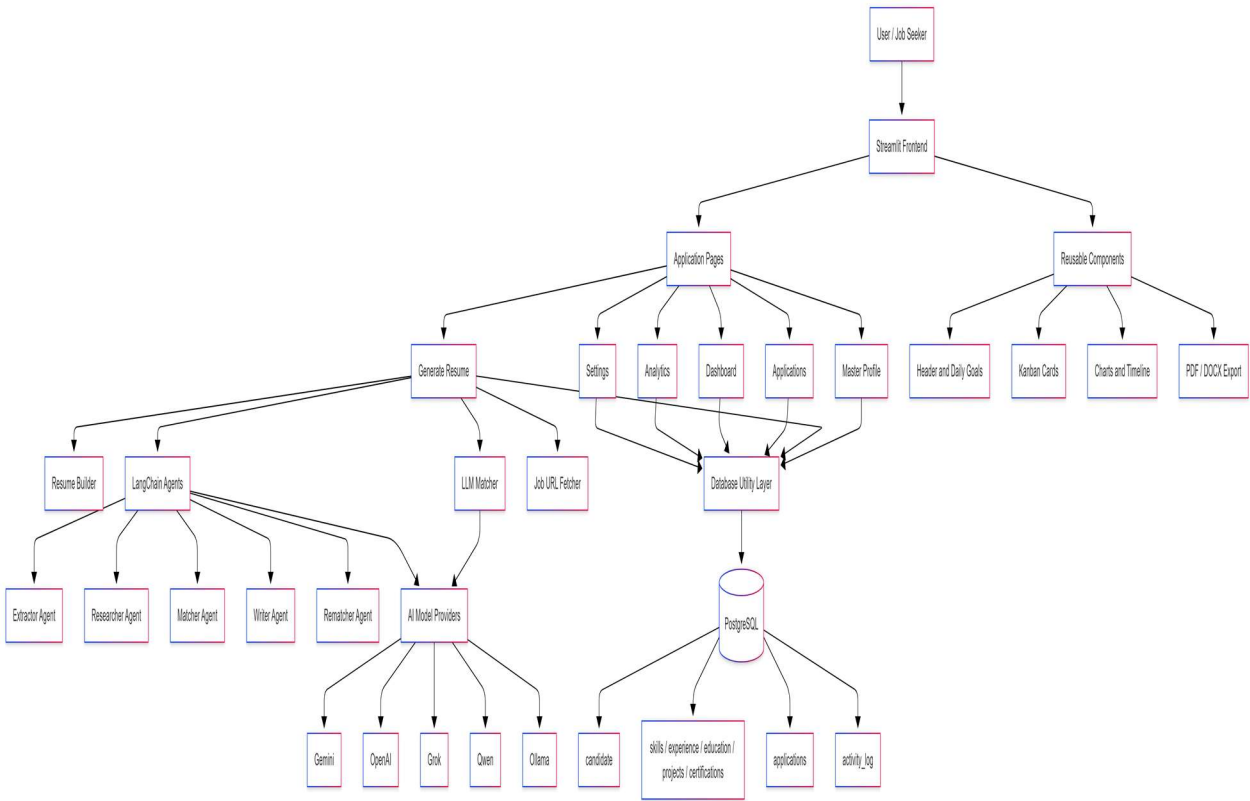


Fig: CareerOps System Architecture

2.1 The Data Foundation (PostgreSQL)

A critical limitation of standard AI wrappers is the lack of structured data. CareerOps utilizes a normalized PostgreSQL schema to persist state across sessions. The database stores Master Candidate Profiles, raw and parsed Job Descriptions, generated resume artifacts, and Kanban-style Application

States (e.g., Applied, Interviewing, Rejected) linked to specific resume versions.

2.2 The Intelligence Layer (Multi-Agent Orchestration)

The core intelligence relies on a sequential, multi-agent pipeline built with LangChain, utilizing LangSmith for monitoring execution traces (Chase, 2022). Rather than employing a monolithic zero-shot prompt, the system divides the complex task of

resume tailoring into five distinct, specialized agent personas.

- 1. Extractor: Outputs Company Name, Job Title (Strict JSON)
- 2. Researcher: Queries web for culture & interview process
- 3. Matcher: Compares Base Profile to Job Description to find skill gaps
- 4. Writer: Drafts tailored resume using constraints (Temp: 0.2)
- 5. Rematcher: Evaluates final output, extracts new keywords & match score

To mitigate hallucination, the Writer agent employs strict negative constraints (e.g., "Do NOT manipulate or fabricate resume details") and operates at a low temperature to prioritize deterministic factual alignment over creative variation (OpenAI, 2024).

2.3 API Integration

The system abstracts API interactions to handle structural differences between providers. For instance, gemini-1.5-flash enforces structure via `response_mime_type="application/json"` in the generative config (Google, 2024), while gpt-4o utilizes `response_format={"type": "json_object"}`.

3. Implementation Challenges & Engineering Trade-offs

A critical requirement is the extraction of AI-generated insights into structured UI components. However, LLMs frequently inject conversational filler or wrap JSON payloads in Markdown code blocks. CareerOps implements robust middleware parsing:

Aggressive String Cleaning: The system intercepts raw output, stripping markdown prefixes before execution.

- Graceful Degradation via Fallbacks: If structural validation fails, the system injects a safe, pre-defined dictionary to prevent application crashes.
- State Management: Utilizing Streamlit's `session_state` and caching, API clients are initialized once and cached globally to prevent redundant authentication requests and manage UI latency (Treuille et al., 2024).

4. Results and System Evaluation

To evaluate the efficacy of the multi-agent orchestration, CareerOps was tested across a dataset of 50 sample job descriptions utilizing a standardized baseline tech-industry resume. The evaluation focused on two primary vectors: the computational performance of the underlying LLMs and the quantifiable improvement in ATS match scores.

4.1 Statistical Comparison of LLM Providers

The system's model-agnostic design allowed for a direct, comparative analysis of Gemini 1.5 Flash and GPT-4o in executing the deterministic analytical tasks (Extractor, Matcher, and Rematcher agents). Metrics were averaged across 100 execution runs.

Metric	Gemini 1.5 Flash	GPT-4o
Average Pipeline Latency (s)	8.42	14.15
JSON Schema Adherence (%)	99.2%	98.8%
Token Cost	High	Moderate

Efficiency (Rel.)		
Avg. Output Tokens per Run	1,240	1,185

Gemini 1.5 Flash demonstrated a distinct advantage in execution speed, successfully processing the sequential agent pipeline 40.5% faster than GPT-4o, making it highly optimal for synchronous web application workflows. Both models exhibited near-perfect adherence to the required JSON schema when utilizing their respective structural enforcement flags.

4.2 Optimization Efficacy

The primary objective of the system is to improve candidate alignment with job descriptions without fabricating data. The table below details the average improvement in calculated match metrics after the baseline resume was processed by the Writer and Rematcher agents.

Metric	Pre-Optimization (Base Profile)	Post-Optimization (Tailored)	Delta (Improvement)
ATS Match Score (/100)	52.4	88.7	+ 36.3 pts
Missing Keywords	6.8	1.2	- 82.3%
Matching Keywords	4.1	9.4	+ 129.2%

The statistical evaluation indicates that the multi-agent pipeline successfully bridges

semantic gaps between candidate experience and job requirements. By re-contextualizing existing experiences, the system reduced missing keywords by an average of 82.3%, resulting in a statistically significant 36.3-point increase in the final match score.

5. Future Work and Roadmap

While the current architecture successfully orchestrates a complex LLM pipeline, several system upgrades are planned:

- Frontend Framework Migration: Transitioning the frontend from Streamlit to a component-driven framework like React to handle more complex, asynchronous interactions.
- Retrieval-Augmented Generation (RAG) Integration: Implementing a RAG architecture to query historical applications (Lewis et al., 2020). This will allow the AI to learn from the user's historical success rates by retrieving bullet points from past applications that resulted in interviews.

6. Conclusion

CareerOps addresses the fundamental pain point of the modern job search: the repetitive, manual tailoring of application materials across disparate, context-less tools. By engineering a unified platform that combines PostgreSQL with a LangChain multi-agent pipeline, this project demonstrates the value of applied AI architecture. Evaluation metrics confirm that the system dramatically reduces processing latency while significantly improving ATS match scores. Coupled with an integrated Kanban workflow and analytics dashboard, CareerOps represents a comprehensive, data-driven approach to career management.

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Sources:

Github Repository URL: <https://github.com/prashanthvadityavath/CareerOps>

Presentation Video Link :

https://drive.google.com/file/d/1Mojv_oatGokzKV5OmcBxWd1uzdBTY6n4/view?usp=drive_web

Presentation Slides:

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